

The Formal Foundation: Concepts from Part 1

The Formal Foundation: Concepts from Part 1 I

Part 3 builds on the formal framework in Part 1. This section summarizes core definitions and theorems using the same notation as the Part 1 manuscript.

The Adversary Hierarchy (Ω)

The Adversary Hierarchy (Ω) I

Part 1 formalized the “Scope of Threat” through a hierarchical taxonomy. This hierarchy allows precise definition of defensive scope.

The Adversary Hierarchy (Ω) I

- ▶ Ω_1 (**External**): The adversary controls inputs (e.g., prompt injection). The agent's internal state is intact.
- ▶ Ω_2 (**Peripheral**): The adversary controls tools or RAG data (e.g., poisoned retrieval). The agent's perception is compromised.
- ▶ Ω_3 (**Agent**): The adversary controls the agent's weights or context (e.g., identity implementation). The agent itself is untrusted.
- ▶ Ω_4 (**Coordination**): The adversary controls a subset of the swarm (e.g., Sybil agents). The consensus mechanism is under attack.
- ▶ Ω_5 (**Systemic**): The adversary controls the orchestrator or infrastructure. The system's rules are compromised.

The Adversary Hierarchy (Ω) I

The simulations in Part 2 show defense difficulty rising non-linearly with this hierarchy: Ω_1 attacks are largely caught by surface-level filters (see Part 2 for category-level rates), while Ω_4 coordination attacks need quorum- and graph-level analysis and remain structurally harder to flag than single-channel injections.

The Trust Calculus (T)

The Trust Calculus (T) I

A central contribution of Part 1 is the **Trust Calculus**, a formal system for reasoning about belief reliability. It defines Trust (T) not as a binary permission but as a continuous property of a belief b , denoted as $T(b) \in [0, 1]$.

The formal definition of Trust Update (Theorem 3.1 in Part 1) establishes that trust must decay across delegation chains:

Theorem 3.1 (Trust Preservation): *For any delegation chain $C = \{a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n\}$, the trust in the final output cannot exceed the trust of the weakest link, degraded by the distance from the source.*

$$T(result) \leq \min_i T(a_i) \cdot \delta^{|C|}$$

where δ is the decay factor ($0 < \delta < 1$).

The Trust Calculus (T) I

This theorem provides the mathematical basis for the “Trust Decay” mechanism evaluated in Part 2. It ensures that uncertainty is preserved and amplified effectively as information travels through the network.

In active inference terms, δ is precision decay: each hop along a delegation chain attenuates channel precision. That matches precision-weighted belief updating—the same idea as weighting sensory evidence by reliability. Part 2 (FEP.1–FEP.2) states the correspondence between CIF’s trust update rules and variational free energy for the shared generative-model setup used there.

The Cognitive Firewall (Φ)

The Cognitive Firewall (Φ) I

The **Cognitive Firewall** is defined in Part 1 as a function Φ that maps inputs to decisions based on three verification layers:

The Cognitive Firewall (Φ) I

1. **Syntactic Verification** (V_{syn}): Checks for structural anomalies.
2. **Semantic Verification** (V_{sem}): Checks for meaning-level violations.
3. **Pragmatic Verification** (V_{prag}): Checks for contextual anomalies.

The Cognitive Firewall (Φ) I

In the Part 2 experiments, this modular structure was shown to be the primary defense against Ω_1 (External) attacks.

The Stealth-Impact Tradeoff

The Stealth-Impact Tradeoff I

Part 1 provides a theoretical bound on attack performance, formalized as the Stealth-Impact Tradeoff.

Stealth-Impact Tradeoff: *For a given defense sensitivity ϵ , the probability of detection $P(d)$ approaches 1 as the divergence of the attack behavior from the baseline increases.*

This formalism suggests that catastrophic attacks are inherently easier to detect than subtle attacks. Part 2's data consistently validated this: High-impact attacks were detected 98% of the time, while low-impact attacks were detected only 74% of the time.

Defense Composition

Finally, Part 1 defines the **Composition Algebra**, determining how output probabilities of distinct modules interact. The key result is that orthogonal defenses compose multiplicatively.

This “Swiss Cheese Model” was supported by Part 2’s parametric simulation, where the full stack reached a 96–100% design-level detection ceiling and outperformed the sum of its parts. The real prototype pipeline is materially lower and is reported separately as a multi-seed mean of approximately 44.8%.

The Science Behind Belief Updates: Free Energy

The Science Behind Belief Updates: Free Energy I

The Cognitive Integrity Framework's formal mechanisms have a deep connection to the **Free Energy Principle (FEP)**—the leading computational theory of how intelligent agents maintain coherent beliefs about their environment [?]. Understanding this connection helps explain *why* CIF's defenses work, not just *that* they work.

What Is Free Energy? I

In computational neuroscience, **variational free energy** F measures how well an agent's internal model Q of the world matches reality:

$$F = D_{\text{KL}}[Q\|P] - \mathbb{E}_Q[\log P(o|s)]$$

The first term penalizes divergence from the prior; the second rewards accurate prediction of observations. Healthy cognition minimizes F —beliefs that minimize free energy are accurate, coherent, and resistant to manipulation.

From the FEP perspective, a **cognitive attack is any intervention that forces an agent's free energy up**. Prompt injections, belief manipulation, and trust exploitation all work by injecting observations (or “observations”—fabricated messages, false context) that drive the agent's belief state Q away from its prior P in ways that serve the attacker's goals rather than the agent's.

Attacks as Free Energy Increases I

CIF formalizes this in Part 2 (FEP.1): an attack ω is detected when $\Delta F(\omega) = F(Q_{\text{attacked}}) - F(Q_{\text{baseline}}) > \kappa_{\text{FEP}}$. The threshold κ_{FEP} is set by the precision of the agent's prior—agents with strong, well-calibrated priors (high precision) require larger perturbations to shift their beliefs, making them more attack-resistant.

Trust as Precision Weighting I

CIF's trust calculus (the δ^d decay) has a natural interpretation under FEP: **trust score = precision weight**. When agent A receives a message from agent B with trust score $T(B \rightarrow A)$, it should weight that message's evidence proportional to $T(B \rightarrow A)$, treating it as an observation with precision $\rho = T(B \rightarrow A)$. Trust decay across delegation chains (δ^d) corresponds to precision attenuation in distal channels—the same mechanism that makes far-away sensory signals less reliable than proximal ones.

The Belief Sandbox as Constrained Inference I

The belief sandbox (Part 1) has a direct FEP interpretation: it is **constrained variational inference** where the update is only accepted if $\Delta F \leq \kappa \cdot \varepsilon_{\text{precision}}$. This is equivalent to requiring that accepted belief updates stay within a bounded geodesic radius on the statistical manifold of belief distributions—exactly Theorem CG.1 from Part 2.

Practical Implication for Operators I

This connection is not just theoretical. It means:

Practical Implication for Operators I

1. **Emergent misalignment is the hardest problem** because it minimizes ΔF per agent: each individual belief shift is sub-threshold, but the collective drift accumulates. This is precisely why colony-scale monitoring is necessary—the FEP signal is distributed across agents.
2. **Trust calibration is precision calibration**: operators who carefully calibrate trust scores are effectively setting the precision weighting of their agent network. Well-calibrated trust $\rightarrow$ robust cognition.
3. **The Ω_5 miss rate (44%) reflects FEP's fundamental challenge**: systematic manipulation by a compromised orchestrator can shift the agent's generative model P itself (not just Q), making the baseline a moving target. This requires out-of-band verification (human review, Byzantine quorum) rather than in-context detection.

Practical Implication for Operators I

For the full mathematical treatment, see Part~2's theoretical-connections and information-geometry sections.

Category-Theoretic Formalization of Defense Composition

Part 2 (§1c, §2c) extends the composition algebra of Part 1 into a full category-theoretic framework. This formalization is relevant to practitioners because it provides *structural guarantees* — not just empirical observations — about how CIF defenses combine.

The Defense Category $\mathcal{D}$ (Part 2, Definition CT.1): The CIF defense suite forms a category whose objects are cognitive states $\sigma \in \Sigma$ and whose morphisms are detection functions $f : \sigma \rightarrow \text{DefenseResult}$. The composition rule formalizes *short-circuit detection*: once any module fires, subsequent modules do not override the event.

Three key theorems (Part 2, Theorems CT.1–CT.3): - **CT.1 (Category Laws)**: Defense composition satisfies identity and associativity — the algebraic scaffold that makes multi-layer defenses predictable. - **CT.2 (Categorical Product)**: Parallel composition is the categorical product in $\mathcal{D}$, with max-score fusion as the universal construction — recovering Part 1's parallel composition rule from first principles. - **CT.3 (Functor Preservation)**: Any defense-preserving transformation (e.g., architecture adapter) that maps morphisms while preserving composition structure cannot reduce the composite detection rate below the guarantee of Theorem 3.1.

Practical value for operators: The categorical framing enables *type-checked composition* (incompatible modules are refused at composition time), *empirical law verification* (the `verify_category_laws()` function in Part 2's `src/formal/category_theory.py` validates the laws against randomly sampled morphism triples), and a unified framework for reasoning about series and parallel configurations. When designing a new CIF deployment, the Defense Category $\mathcal{D}$ provides the structural language to specify *what it means* for two defense modules to compose correctly.

Part 2 also provides a **composable visualization engine** (DefenseGraph, CategoryDiagram, LatticeViz, OperadPlot, MonadFlow, LensDiagram) and an **interactive CIF Composer web UI** (output/web/cif_composer.html) — a self-contained HTML/JS/D3 application with 8-module palette, live metric computation based on Theorems 3.1/3.2, category law verification, 4 deployment presets, and Python/JSON/SVG export. Operators can use the Composer to explore deployment configurations before committing to a production layout.